EWB Edutech Private Limited

Amazon Web Services (AWS) Cloud Computing Internship

TECHNICAL PROJECT REPORT

AWS VPC PEERING AND SECURE ARCHITECTURE IMPLEMENTATION

Submitted By:

Name: Nithyasri K S

Degree & Branch: B.Tech - CSE

Role: AWS Cloud Computing Intern

Date of Submission: October 24, 2026

1. Executive Summary

- **Objective:**
  The primary objective of this project was to design, deploy, and evaluate an enterprise-grade AWS multi-VPC networking architecture utilizing non-overlapping IP address spaces (`10.10.0.0/16` and `10.20.0.0/16`). The implementation focused on establishing private, low-latency inter-VPC communication over a dedicated AWS VPC Peering connection without exposing internal network traffic to the public internet. To enforce strict perimeter defenses, private application workloads were isolated from direct external ingress, leveraging a single public Bastion jump-host for secure administrative access. Finally, cross-VPC routing tables, security group rules, and SSH Agent Forwarding were end-to-end validated to ensure robust zero-trust network isolation.

- **Key Architecture Features:**
  - Dual VPC setup with non-overlapping CIDR blocks (`10.10.0.0/16` and `10.20.0.0/16`).
  - Restricted internet entry point via a public Bastion host in VPC A.
  - Private workloads in `VPC A` and `VPC B` completely isolated from public internet exposure (no public IPs assigned).
  - Direct inter-VPC traffic routing using an active AWS VPC Peering connection (`ewb-pcx-a-b`).
  - **Granular Least-Privilege Security Groups:** Inbound security group rules restricted SSH and ICMP traffic strictly to explicit private `/32` host IP addresses across VPC boundaries.
  - **Keyless Jump-Host Authentication:** Implemented SSH Agent Forwarding to securely transit through the Bastion host to private workloads without storing private key files (`.pem`) on intermediate servers.

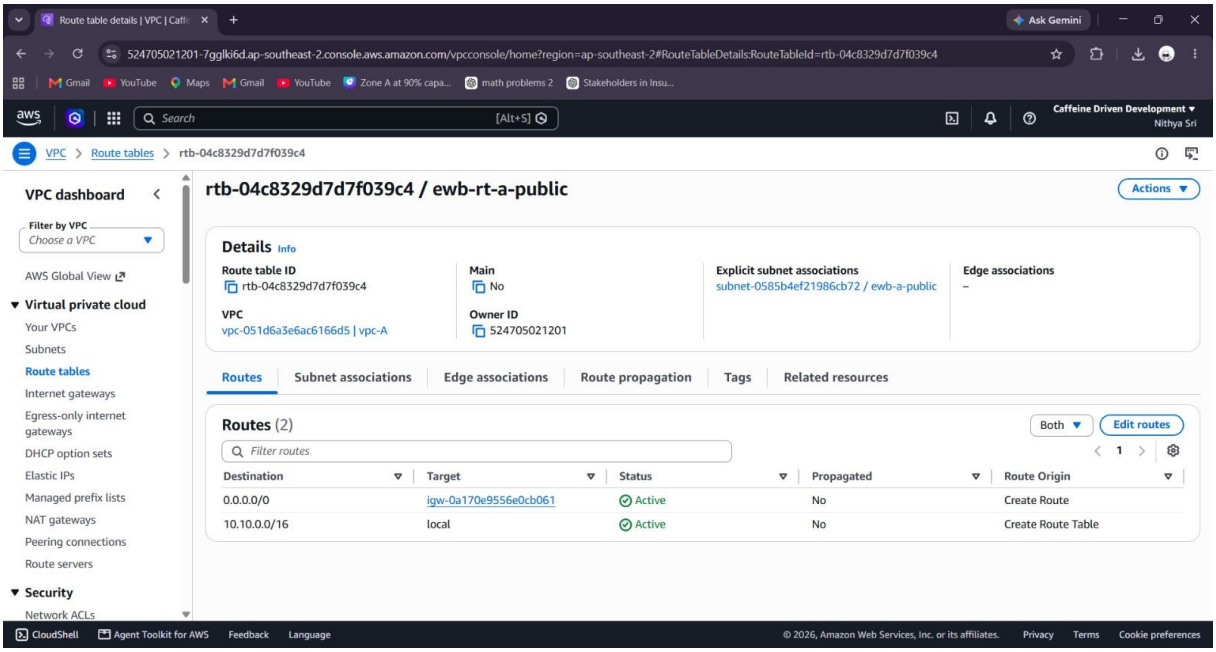
2. Infrastructure & Address Allocation Table

Component	Resource ID / Name	VPC	Subnet / CIDR	IP Address Allocation
VPC A	`ewb-vpc-a`	`ewb-vpc-a`	`10.10.0.0/16`	Primary VPC
VPC B	`ewb-vpc-b`	`ewb-vpc-b`	`10.20.0.0/16`	Peered Target VPC
Public Subnet	`ewb-a-public`	`ewb-vpc-a`	`10.10.1.0/24`	`ap-southeast-2a`
Private Subnet A	`ewb-a-private`	`ewb-vpc-a`	`10.10.2.0/24`	`ap-southeast-2a`
Private Subnet B	`ewb-b-private`	`ewb-vpc-b`	`10.20.1.0/24`	`ap-southeast-2a`
Bastion Host	`ewb-bastion`	`ewb-vpc-a`	`ewb-a-public`	Public IP: `32.236.163.27` / Private: `10.10.1.10`
App Instance A	`ewb-app-a`	`ewb-vpc-a`	`ewb-a-private`	Private IP: `10.10.2.10` (No Public IP)
App Instance B	`ewb-app-b`	`ewb-vpc-b`	`ewb-b-private`	Private IP: `10.20.1.10` (No Public IP)
VPC Peering	`ewb-pcx-a-b`	VPC A $\leftrightarrow$ VPC B	N/A	Active Status

3. Implementation & Evidence (Checkpoints)

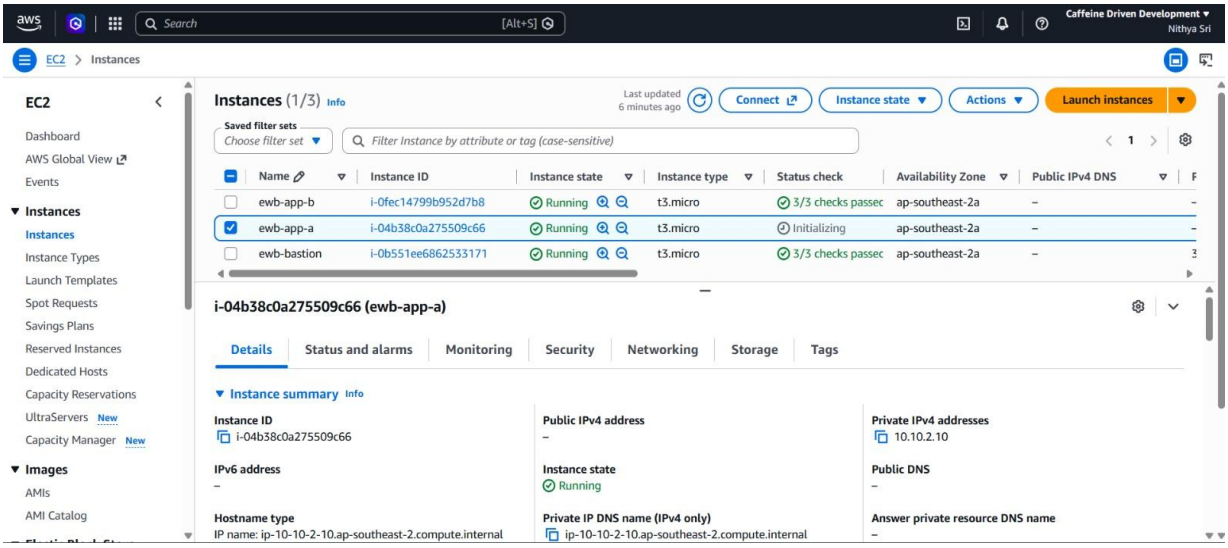
Checkpoint 01: Gateway & Public Route Configuration

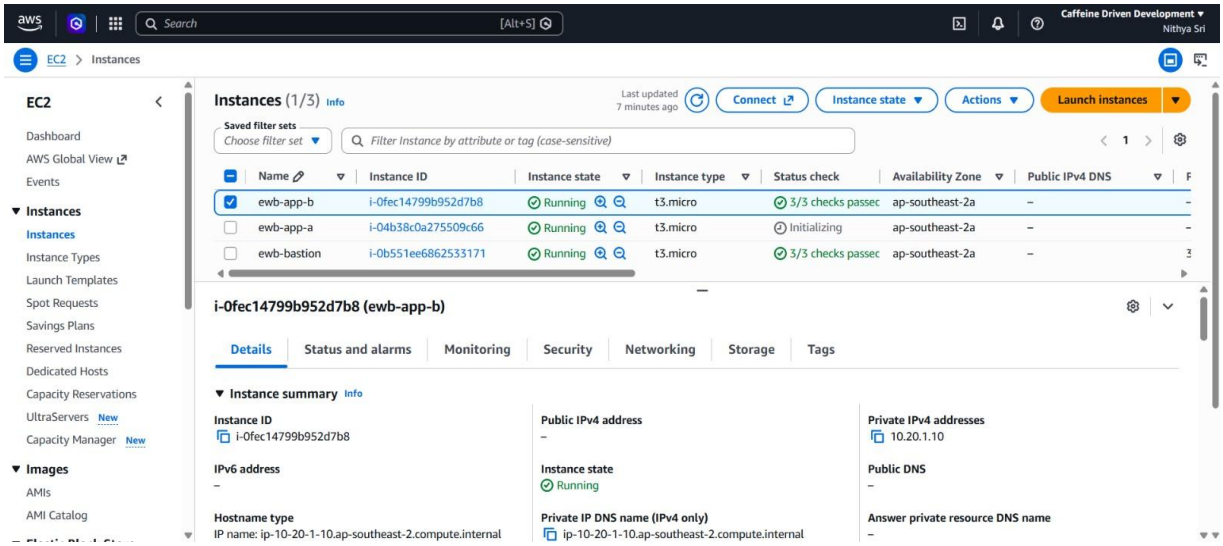
- **Details:** Provisioned `ewb-igw-a` attached to `ewb-vpc-a`. Configured route table `ewb-rt-a-public` with a default destination route (`0.0.0.0/0`) targeting the Internet Gateway, granting the Bastion host external internet reachability.



Checkpoint 02: Network Isolation & Private Workload Deployment

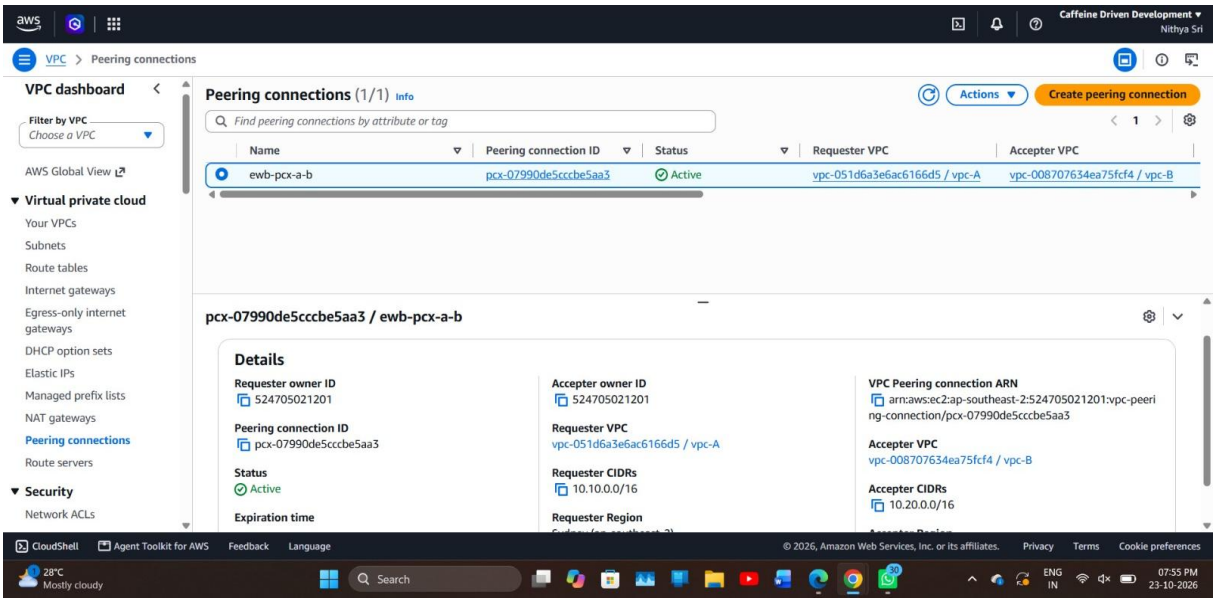
- **Details:** Deployed `ewb-app-a` and `ewb-app-b` into isolated private subnets. Verified that auto-assignment of public IP addresses was disabled, ensuring zero direct ingress/egress internet connectivity for application servers.





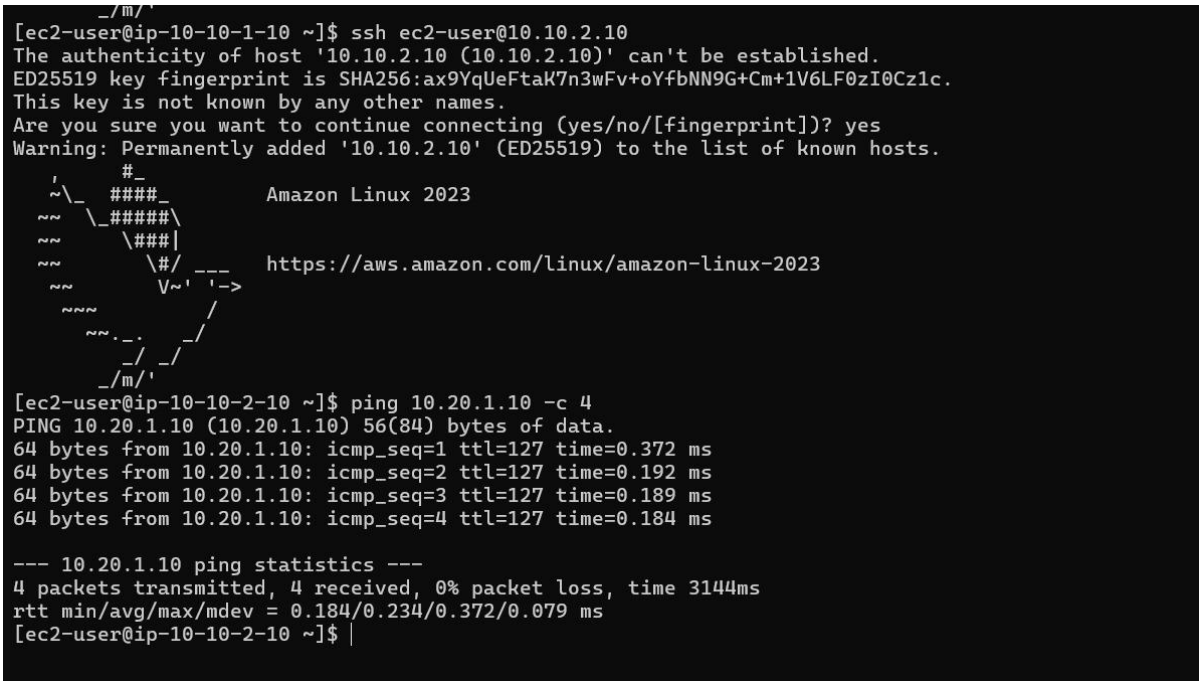
Checkpoint 03: VPC Peering Establishment & Route Table Updating

- **Details:** Established active peering link `ewb-pcx-a-b` between Requester `10.10.0.0/16` and Accepter `10.20.0.0/16`. Added target routes across `ewb-rt-a-public`, `ewb-rt-a-private`, and `ewb-rt-b-private` to direct cross-VPC traffic over the peering pipeline.



Checkpoint 04: End-to-End Private Connectivity & Security Verification

- **Details:** Configured local SSH Agent Forwarding to authenticate through `ewb-bastion` (`32.236.163.27`) and connect to `ewb-app-a` (`10.10.2.10`). Executed ICMP connectivity testing from `ewb-app-a` to `ewb-app-b` (`10.20.1.10`), achieving 100% packet transmission success across the VPC Peering link with 0% loss.



4. Technical Conclusion & Security Takeaways

- **Private Network Perimeter:** Workloads in separate VPCs communicate securely across AWS backbone infrastructure without traversing the public internet.
- **Identity & Access Management:** Security Groups restricted inbound ICMP and SSH access specifically to authorized private IP addresses, enforcing least-privilege network security principles.